

Lesson 6-4 · Quick

Logarithmic Functions + DOK-3 Sales Revenue Inverse (Item 28)

Klimesara-adapted · Student-centered · No Blooket

DOK 1-2

7 min

bridge to log inverses

Three warm-up items (solo, 7 min)

1. **q31** — Translation multi-select for $h(x) = \ln(x + 2) - 1$. Which statements are true?
2. **q5** — Graph $y = \log_4 x$. State domain, range, VA, x-intercept.
3. **q32 (SAT/ACT)** — Find the inverse of $f(x) = 5^{x+1}$. (Choices A–D.)

DOK 2

8 min

teacher models

Rules Card — Log Key Features + Inverse Rules (keep on your page) Key features of $y = \log_b x$: Domain $x > 0$; VA: $x = 0$; x-int: $(1, 0)$; passes through $(b, 1)$.

Translated form $y = \log_b(x - h) + k$: VA shifts to $x = h$; domain $x > h$.

Inverse rules:

- $y = a \cdot b^x \Rightarrow f^{-1}(x) = \log_b(x/a)$
- $y = \log_b(x - h) + k \Rightarrow f^{-1}(x) = b^{(x-k)} + h$

Composition check: $f(f^{-1}(x)) = x$

DOK 2

8 min

teacher models

Ex 3B: Find the inverse of $g(x) = \log_7(x + 5)$

1. Replace $g(x)$ with y : $y = \log_7(x + 5)$
2. Swap x and y : $x = \log_7(y + 5)$
3. Rewrite in exponential form: $7^x = y + 5$
4. Solve for y : $y = 7^x - 5 \Rightarrow g^{-1}(x) = 7^x - 5$
5. **Composition check:** $g(g^{-1}(x)) = \log_7(7^x - 5 + 5) = \log_7 7^x = x \checkmark$

90-sec contrast — Ex 3A: inverse of $y = 3 \cdot 5^x$: same swap-and-solve, direction reversed
 $\rightarrow y = \log_5(x/3)$.

Explore — Think-Pair-Share (25 min)

Algebra 2 · Lesson 6-4

DOK 2–3

25 min

student work

5 items in order. Plan ~10 min for Practice #28 (DOK-3). 45-min Wed F: skip q9.

Practice #28 · ★ DOK-3 SALES REVENUE INVERSE Sales revenue model. Find the inverse. Judge which equation form is easier for a given R .

Practice #21 · Inverse of $f(x) = 5^{x-3}$ Find the inverse. [Form B Q7 rehearsal]

Practice #24 · Inverse of $f(x) = \log_2(8x)$ Swap and rewrite. Watch for domain. [Form B Q12 rehearsal]

DOK 2

5 min

DOK-3 reveal + 6-5 bridge

Sentence frame · Practice #28 “For Practice #28, I found the inverse by _____.
For revenue $R =$ _____ dollars, the original form is easier because _____.
The inverse form is easier when _____.”

Bridge to Lesson 6-5 Log properties (product, quotient, power rules) let you expand or condense log expressions — making inverses and equations faster.

DOK 1–2

5 min

not graded

Complete on your exit ticket

1. Find the inverse of $f(x) = 3 \cdot \log_5(x - 2) + 1$.
2. State the vertical asymptote of f .
3. State the horizontal asymptote of f^{-1} .
4. One biggest thing I learned today: _____.